

Challenge

```
void main() {  
    if (z == 1)  
        then if (x == 1)  
            then y = 1  
            else y = 0  
        else x = y  
}
```

Analyze ***non-interference***

Changing values of higher-level variables does not impact values of lower-level variables.

The technique should be: sound, static, compositional, automatic, practical.

Information Flow Calculus

```
void main() {  
    if (z == 1)  
        then if (x == 1)  
            then y = 1  
            else y = 0  
        else x = y  
}
```

Label in and out variables

Example: $x = \text{exp}$

$$\text{In}(C) = \text{Occ}(\text{exp})$$
$$\text{Out}(C) = x$$

Record data flows in a matrix.

Security Flow Matrix

$$\begin{array}{c} \begin{array}{ccc} & x & y & z \\ \begin{array}{c} x \\ y \\ z \end{array} & \begin{pmatrix} \cdot & \text{💧} & \cdot \\ \text{💧} & \cdot & \cdot \\ \text{💧} & \text{💧} & \cdot \end{pmatrix} \end{array}$$

- no dependency from v_{in} to v_{out}
- 💧 NI violation if $\ell(v_{\text{in}}) \not\leq \ell(v_{\text{out}})$

Evaluate satisfiability

- Constraints from matrix
- Information flow policy
- Variable security classes

SAT means program is non-interfering.